CONSTITUTIONAL INTELLIGENCE SERIES · PAPER 13 · CONSOLIDATION

The Claim Architecture

Patent Families Over the Formal Kernel

The consolidation layer, the eleven-paper formal architecture compressed into eight foundational invention families: the object of invention restated as the computational governance of authority under uncertainty, the technical-mechanism test that separates claim from principle, the dependency graph and continuation strategy for the clinical application families, the benchmark as utility evidence, the eighteen-paper scientific portfolio, and the defensive-publication doctrine that keeps the field open while the mechanisms are protected

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Compiled under: the reconciled SAVANT PRIME architecture · ARCHITECT persona

Version 1.0 · 2026-10-06

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0.1 Provenance and Position in the Program

This is Paper 13, the consolidation paper. Papers 0-10 discharged the eleven-installment program defined by the SVT Architect Analysis file, the kernel, its algebra, its composition laws, its negative space, its metacognition, its epistemic hygiene, its objective transitions, its survivability, its feasibility pricing, its generative lifecycle, and its domain modules. But the source does not end at the eleventh specification. After the paper sequence it states a further instruction in two movements: first, that the eighteen candidate invention families of its opening analysis reduce to approximately eight foundational patent families, named A through H; second, that the clinical implementations become continuation, divisional, and application opportunities, subject to prior-art and patent-eligibility analysis VERIFIED-FROM-SOURCE. This paper formalizes that reduction. It is the program's answer to the question the source poses in its first paragraph: what, precisely, is the patentable surface of an architecture whose intellectual center is not any clinical protocol but the control machinery that governs machine authority itself VERIFIED-FROM-SOURCE.

Three interpretive decisions are recorded here and defended in the body. First, the source presents the eight families twice with different boundaries, an earlier enumeration whose family C fused the authority lifecycle with precedence, and a later, explicitly terminal enumeration that separates them and absorbs resource-constrained optimization into the runtime's admissibility machinery. The later enumeration governs; the refinement is recorded as decision D-12.2 in §17 FORMALIZED-HEREIN. Second, the source's own legal discipline is adopted verbatim as this paper's epistemic law: every family is a candidate invention family, not a claim of legally established novelty; patentability requires prior-art searching, claim construction, and jurisdiction-specific examination VERIFIED-FROM-SOURCE. Every legal-effect statement in this document is therefore tagged UNKNOWN or CONJECTURE; only structure is formalized here. Third, the source's allocation doctrine, patent only where the implementation creates a concrete technical mechanism; publish foundational principles defensively, is elevated from advice to a computable test, the Technical Mechanism Test of §12 FORMALIZED-HEREIN.

0.2 Notation

All notation of Papers 0-10 is retained. New here: FAMX, the family object for $X \in \{A,\ldots,H\}$, with the six-slot schema of §3; mech(c), the mechanism predicate: claim c recites a concrete, executable mechanism; TMT, the Technical Mechanism Test of §12; $c \lhd c'$, claim dependency: c narrows c′; Found / App, the foundational and application claim sets; Objects(C1..11), the discharged formal objects of the eleven predecessor papers. Load-bearing inheritances: Paper 0's kernel and authority algebra; Paper 8's admissibility fixed point $A_{adm} = \mathrm{gfp}(\Phi_R)$; Paper 9's lifecycle automaton Σ_G; Paper 10's compilation predicate compat(m, K) and registry.

0.3 Epistemic status convention

Every claim carries one of four tags, as fixed in Paper 0 and enforced throughout: VERIFIED-FROM-SOURCE for statements present in the source architecture; FORMALIZED-HEREIN for precise statements constructed in this document from source intent; CONJECTURE for hypotheses the architecture suggests but does not yet entail; UNKNOWN for open questions stated without a proposed answer. In this paper the convention carries an extra load: no statement about legal validity, novelty, non-obviousness, enablement, or freedom-to-operate rises above CONJECTURE, and most stand at UNKNOWN, because the searches and examinations on which they depend have not been performed. The falsification register in §16 lists the conditions under which each load-bearing structural claim would fail; §18 records the outstanding proof obligation PO-12.1. Nothing is asserted beyond its tag.

0.4 Authority boundary

C0 informational. This document is legal-adjacent: it structures candidate claims, it files none, and it is not legal advice. Every claim sketch herein is a formal specification of what would be claimed, drafted to make prior-art searching maximally efficient, not a representation that any claim is novel, enabled, or eligible in any jurisdiction. The architecture's own discipline applies to its author: authority here is zero; the document's only powers are description, mapping, and the precise statement of obligations owed to patent counsel and to the scientific record.

0.5 The thesis in one sentence

The invention is not any protocol but the transition itself, capability becoming constitution,

constitution becoming authority, authority becoming action, action becoming verification, verification becoming learning, and the claim architecture is the eleven-paper kernel read once more, this time as law.

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Part I: The Object of Invention

Before anything can be claimed, the architecture must say what it is an invention of. The source answers with unusual precision: not a clinical protocol, not a prompt library, but the computational governance of authority under uncertainty, and it compresses that answer into a single six-stage arrow.

1 The Compression Spine

The source closes its architectural statement with a boxed equation it calls the decisive architectural compression VERIFIED-FROM-SOURCE:

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Capability → Constitution → Authority Eq 1-1
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Authority → Action → Verification → Learning Eq 1-2 The source presents the spine as one six-term arrow; it is set here as two display lines solely for the page measure. The arrow is not a pipeline of components but a chain of entitlements: each stage is a gate the next stage may not bypass, with failure, uncertainty, mortality, scarcity, and environmental degradation acting on authority rather than appended as warnings VERIFIED-FROM-SOURCE.

Read against the eleven predecessors, every arrow is already a theorem-shaped object. Capability is the model's raw competence, deliberately left ungoverned, because the architecture's first law is that capability confers no authority (Paper 0). Constitution is the kernel K with its invariants (Papers 0, 2). Authority is the dynamically computable quantity At of

the authority algebra (Papers 0, 1). Action is the execution of an admissible proposal, Aadm, the greatest fixed point of the feasibility operator ΦR (Paper 8). Verification is the audit-and-hygiene machinery that prices evidence and decays standing (Paper 5). Learning is the governed revision loop, protocol lifecycle, mortality-attributed updates, registry memory (Papers 6, 9, 10) FORMALIZED-HEREIN. The spine is therefore not a slogan over the architecture; it is the architecture's table of contents written as a single arrow, and every claim sketched in Part II will cite its stages as elements.

DEFINITION 1.1: THE OBJECT OF INVENTION

The object of invention is the computational governance of authority under uncertainty: the machinery that determines under what conditions an intelligent system is entitled to act, how much authority it may exercise, when that authority must contract, who or what can revoke it, and what evidence is required before authority is restored. The fundamental primitive is the Authority State, not the prompt: every component compiles into a state-bearing object carrying objective, world-state, evidence state, uncertainty state, authority level, permitted and prohibited actions, verification obligations, escalation, expiry, withdrawal, recovery conditions, and audit lineage. VERIFIED-FROM-SOURCE, the source states both the object ("the computational governance of authority under uncertainty") and the primitive ("The fundamental primitive should be Authority State, not Prompt") in its opening analysis. Paper 0 formalized the object; this paper claims it.

One proposition of the source deserves to stand at the head of this paper because it is the claim architecture's conscience: a sufficiently capable intelligence must possess mechanisms for reducing its own effective authority faster than it can increase its capability when uncertainty, objective instability, environmental degradation, or legitimacy failure rises VERIFIED-FROM-SOURCE. Formally, with capability κ and effective authority Aeff both functions of time:

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dAeff/dt < 0 whenever dκ/dt ≥ 0 ∧ (U ↑ ∨ O unstable ∨ Eenv ↓ ∨ L ↓)          Eq 1-3
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Papers 1, 3, 6, and 8 each discharged one disjunct of this antecedent, escalation on uncertainty, negative-space contraction, objective-guard transitions, degradation descent. The claim architecture's first structural decision follows immediately: the independent claims of Part II are drafted so that this contraction law is an element, not a preamble, because a principle recited as aspiration is unprotectable and unenforceable, while a mechanism recited as a state transition is both testable and, if novel, claimable FORMALIZED-HEREIN.

2 The Hierarchy Doctrine and the Taxonomy Refinement

The source's portfolio instruction is hierarchical before it is enumerative: the strongest patent portfolio should be hierarchical rather than twenty unrelated inventions, with foundational families first and domain implementations, pediatrics, maternal health, emergency/trauma, palliative care, AMR, as application families VERIFIED-FROM-SOURCE. The reason is structural, not merely strategic. Paper 10 established that the five clinical deltas are readings of one kernel; if each were filed as an unrelated invention, the claim set would assert five disjoint mechanisms where the architecture asserts one, a falsification of the architecture by its own legal record. The hierarchy doctrine is therefore the compilation contract restated in claim space: one independent mechanism, many dependent readings FORMALIZED-HEREIN.

Table 2-1: The taxonomy refinement: earlier enumeration vs. terminal enumeration

EARLIER ENUMERATION (SOURCE, FIRST PASS)	TERMINAL ENUMERATION (SOURCE, FINAL)	REFINEMENT
(A) Constitutional Runtime / Machine Authority Kernel	A. Constitutional Intelligence Runtime	Scope narrowed to the runtime; authority lifecycle extracted to B
(B) Negative-Space Governance Engine	D. Negative-Space / Failure-Boundary Engine	Renamed and reordered; failure boundary made explicit
(C) Authority Lifecycle + Precedence Engine	B. Dynamic Machine Authority Engine; C. Constitutional Precedence Engine	Split: lifecycle (B) separated from conflict resolution (C)
(D) Bounded Epistemic Audit	E. Epistemic Authority Engine	Generalized from audit to evidence-authority coupling
(E) Objective/Mortality Transition Engine	F. Objective Transition Engine	Mortality folded into objective legitimacy (Paper 6's reading)
(F) Graceful Degradation/Continuity Engine	G. Resilient Cognitive Degradation Engine	Renamed; invariant-preserving descent emphasized
(G) Resource	(absorbed into A)	Absorbed; feasibility pricing is kernel

(G) Resource-Constrained Clinical Optimization	(absorbed into A)	Absorbed: feasibility pricing is kernel admissibility (Paper 8's ΦR), not a separate engine
(H) Generative Protocol + Meta-Ablation Engine	H. Constitutional Protocol Lifecycle Engine	Unified as one lifecycle: generation → validation → deployment → monitoring → revision → ablation

The terminal enumeration governs, and the differences are recorded as decision D-12.2 (§17) under the program's standing rule that the stricter, later constraint wins. Three of the refinements deserve defense here rather than mere registration. The B/C split is forced by Paper 1 and Paper 2 being different mathematical objects: the authority lifecycle is a state machine over one agent's authority; precedence is an operator over simultaneous constitutions. A fused family would claim a disjunction, and a disjunctive independent claim is exactly the kind of overreach this architecture exists to refuse, in its own legal record as everywhere else. The absorption of resource-constrained optimization into A is forced by Paper 8's central result: feasibility is not an engine beside admissibility but admissibility itself, computed under an endowment. To claim it separately would be to claim the same fixed point twice. The unification of H is forced by Paper 9's lifecycle automaton ΣG: generation and ablation are the same state's endpoints, and the source's terminal naming, generation → validation → deployment → monitoring → revision → ablation, is precisely ΣG's transition ordering FORMALIZED-HEREIN.

INVARIANT I-HIER: THE HIERARCHY INVARIANT

Every claim in the portfolio is either (i) an independent claim of exactly one foundational family FAMX, X ∈ {A,...,H}; or (ii) a dependent claim narrowing such an independent claim; or (iii) an application-family claim that depends, through the module registry of Paper 10, on at least one foundational independent claim. No application claim asserts a mechanism not present in a foundational family. FORMALIZED-HEREIN, formalization of the source's hierarchy doctrine. Falsification: an application module whose dossier requires a mechanism with no foundational home; H80 quantifies.

What the hierarchy purchases is coherence under examination. When an examiner, a court, or a reviewer asks what the invention is, the portfolio answers with one spine (Eq 1-1/1-2), eight engines, and a registry of readings, and every answer at every level cites the level above it. That is the same property Paper 2 demanded of constitutions and Paper 10 of modules; the claim architecture inherits it by construction.

Part II: The Eight Foundational Families

Each family is specified once, in one schema: the source's definition, the formal machinery of Papers 0-10 that the family claims, the concrete technical mechanism that survives the Technical Mechanism Test, an independent-claim sketch, dependent directions, and the family's exposure to prior art. The sketches are specifications of what would be claimed, drafted to make the prior-art search efficient, not to assert its outcome.

3 The Family Schema

A patent family, treated with the same discipline the architecture applies to modules, is a six-slot object. The schema below is this paper's analogue of Paper 10's M-OBJ: just as no module runs without its nine slots discharged, no family enters the portfolio without all six slots filled, and slot six, the prior-art exposure, begins at UNKNOWN for every family by law of this paper.

Table 3-1: FAM-OBJ: the six-slot family schema

SLOT	NAME	CONTENTS	FAILURE MODE OF OMISSION
FAM.1	Source definition	The terminal enumeration's one-paragraph charter, quoted	The family drifts from the architecture it claims
FAM.2	Formal machinery	The discharged objects of Papers 0-10 the family reads on: invariants, operators, automata, fixed points	The claim floats free of the formalism, unenableable on its face
FAM.3	Technical mechanism	The concrete, executable core: data structures, state transitions, executable steps, what survives §12's TMT	The claim recites a principle; §12 routes it to defensive publication
FAM.4	Independent-claim sketch	One structural sentence: a system comprising..., elements keyed to the spine of Eq 1-1/1-2	The family has no center; dependents orbit nothing
FAM.5	Dependent directions	The enumerated narrowings: parameterizations, domain instantiations, composition rules	Fallback positions missing at examination
FAM.6	Prior-art	The densest adjacent literature, named as search	Surprise at examination; the

FAM.6	Prior-art exposure	The densest adjacent literature, named as search targets, never as conclusions	Surprise at examination; the unsearched family is the unowned family

Two schema laws mirror Paper 10's dossier invariants. First, the schema is complete or the family is inert: a family with an empty FAM.3 slot does not proceed to drafting, it proceeds to the defensive-publication ledger of §12, because the architecture will not claim what it cannot ground in a mechanism. Second, every slot cites upward: FAM.2 cites papers, FAM.4 cites FAM.3, FAM.5 cites FAM.4, and §13's dependency graph cites FAM.4 for every application claim. The claim architecture is a governed object of the same shape as everything else the program has built FORMALIZED-HEREIN.

4 Family A: Constitutional Intelligence Runtime

FAM.1, source definition. "The executable governance kernel" VERIFIED-FROM-SOURCE. Family A is the platform invention: the runtime that sits between an intelligent model and consequential action, so that the model proposes and the kernel determines whether the proposal is admissible VERIFIED-FROM-SOURCE.

FAM.2, formal machinery. Paper 0's kernel K and its invariant set Σ; Paper 8's admissibility fixed point Aadm = gfp(ΦR) with the feasibility structure R = ⟨C, F, τ, E⟩; Paper 10's compilation predicate compat(m, K) and its invariant I-MC (no runtime without compilation); the registry with its measured dominance rule. Family A absorbs the source's first-pass family (G), resource-constrained optimization, because Paper 8 proved feasibility is admissibility under endowment, i.e., kernel behavior, not an auxiliary engine (§2, D-12.2).

FAM.3, technical mechanism. A runtime interposition layer with three executable components: (i) an admissibility evaluator that computes, for each model proposal p, whether $p \in Aadm$ under the current endowment and invariant set, a monotone operator iteration with the termination certificate of Paper 8; (ii) a compilation gate that admits a module to execution only on a discharged nine-slot checklist (Paper 10's COMPILE pipeline); (iii) a state store in which every governed object carries its authority state and audit lineage as mutable, machine-checked fields. All three are data-structure-plus-transition specifications; none is a statement of purpose FORMALIZED-HEREIN.

INDEPENDENT-CLAIM SKETCH A-1

A system comprising: a model interface receiving candidate actions from an intelligent model; an admissibility evaluator computing, for each candidate action, membership in a fixed-point admissibility set defined over resource, feasibility, temporal, and evidence constraints; a compilation gate storing, for each executable module, a nine-slot discharge record and blocking execution absent discharge; and a state store maintaining for each governed object an authority state and an append-only audit lineage, wherein no candidate action executes unless the evaluator certifies admissibility and the acting module's discharge record is current. FORMALIZED-HEREIN, structural sketch; novelty and eligibility UNKNOWN pending PO-12.1.

FAM.5, dependent directions. The runtime parameterized by: evidence-tier pricing (Paper 5); degradation envelopes (Paper 7); endowment families with default floors and price-of-infeasibility routing (Paper 8); module registries with measured dominance (Paper 10); cross-domain instantiations, clinical, financial, infrastructural, under the same element structure (H83).

FAM.6, prior-art exposure. Densest neighbors: agent-orchestration runtimes, policy engines and attribute-based access control, guardrail middleware, safety-interlock architectures in avionics and process control. Search targets are named in B-47. Exposure assessment:

UNKNOWN, the neighbors are adjacent in function but, on the architecture's own analysis, none computes admissibility as a fixed point over a constitutional invariant set with a compilation gate; whether that difference survives search is precisely what PO-12.1 must answer.

5 Family B: Dynamic Machine Authority Engine

FAM.1, source definition. "Authority acquisition, limitation, transfer, revocation, recovery" VERIFIED-FROM-SOURCE. The source's seven-layer program fixes the algebra: authority at time t is a function of evidence, uncertainty, system/world state, objective legitimacy, resource constraints, institutional legitimacy, and verification capacity, dynamically computable rather than binary VERIFIED-FROM-SOURCE:

At = f (Et, Ut, St, Ot, Rt, Lt, Vt)

Eq 5-1

FAM.2, formal machinery. Paper 1's authority algebra in full: the operators grant, constrain, transfer, suspend, revoke, recover, expire; the authority state machine with its proven

unreachable transitions; the contraction law of Eq 1-3 as an executable transition rule; Paper 4's escalation triggers as authority-contracting events; Paper 5's standing-decay as automatic authority erosion. The lifecycle's endpoint discipline, authority does not leak across revocation, and recovery re-enters through re-authorization, never through habit, is Paper 1's contribution and this family's spine.

FAM.3, technical mechanism. (i) An authority register per governed object: a typed tuple ⟨level, scope, evidence floor, expiry, verification obligations⟩ updated only by the seven operators; (ii) a transition guard evaluating each operator's preconditions against the current evidence and uncertainty state, with prohibited transitions rejected by construction (Paper 1's unreachability results made executable); (iii) a decay daemon applying standing-decay as time-indexed authority erosion with audit events. The mechanism's signature: authority is a stored, typed, expiring quantity, never an attribute of the model FORMALIZED-HEREIN.

INDEPENDENT-CLAIM SKETCH B-1

A system comprising: an authority register storing, for each governed object, an authority level, scope, evidence floor, expiry condition, and verification obligations; a transition guard implementing exactly seven authority operators, grant, constrain, transfer, suspend, revoke, recover, expire, each with executable preconditions over evidence quality, uncertainty, and system state; and a decay process that erodes stored authority as a function of elapsed time and evidence staleness, wherein authority decreases monotonically whenever uncertainty rises, objective legitimacy destabilizes, environmental integrity degrades, or institutional legitimacy lapses, independent of any change in model capability. FORMALIZED-HEREIN, structural sketch; novelty and eligibility UNKNOWN pending PO-12.1.

FAM.5, dependent directions. Authority scopes keyed to objective spaces (Paper 6); handoff protocols to human adjudication with re-authorization thresholds; multi-agent authority transfer with lineage preservation; expiry schedules keyed to evidence half-lives (Paper 5).

FAM.6, prior-art exposure. Densest neighbors: capability-based security, delegation and credential frameworks (SPKI/SDSI-style), RBAC/ABAC revocation, OAuth-style token expiry, human-in-the-loop escalation systems. This family and Family E carry the architecture's heaviest exposure (H84): the vocabulary of grant/revoke/expire is sixty years old. The candidate distinction is the precondition structure, evidence-quality-and uncertainty-conditioned transitions with capability-independent contraction, but the search, not the architecture, decides UNKNOWN.

6 Family C: Constitutional Precedence Engine

FAM.1, source definition. "Conflict resolution among simultaneously active constraints" VERIFIED-FROM-SOURCE. The source flags this as potentially one of the most important inventions in the entire analysis: the engine that converts a collection of protocols into an actual governed system when pediatric safety, emergency medicine, maternal autonomy, palliative objectives, resource scarcity, and infrastructure collapse are all active simultaneously VERIFIED-FROM-SOURCE.

FAM.2, formal machinery. Paper 2's composition operators and conflict operators; the precedence order over constraint classes with its proven resolution properties; Paper 10 §4's meet-union composition with the inter-module externality ledger and the measured dominance form $v\sigma(m) \leq v\sigma(K) + \varepsilon m$; the E-protocol for cross-module externalities.

FAM.3, technical mechanism. (i) A constraint classifier assigning every active constraint to a typed class with a declared precedence tier; (ii) a conflict detector computing pairwise and n-ary incompatibilities among simultaneously active constraints over a shared proposal; (iii) a resolution executor applying the precedence operator to produce a single admissible set, with meet-union semantics for module composition and the externality ledger consulted on every cross-module resolution; (iv) a resolution log recording every conflict, its participants, and its resolution, because a precedence engine that cannot explain itself is, by Paper 5's law, an authority without standing FORMALIZED-HEREIN.

INDEPENDENT-CLAIM SKETCH C-1

A system comprising: a classifier assigning each active constraint to a precedence tier within a declared constitutional order; a conflict detector computing incompatibilities among simultaneously active constraints bearing on a common candidate action; and a resolution executor that computes the admissible action set as the meet of the active constraints under the precedence order, records each conflict and its resolution to an append-only log, and routes cross-module externalities to a shared ledger whose entries constrain subsequent resolutions, wherein the resolution is total: every conflict among active constraints yields either an admissible set or a certified refusal. FORMALIZED-HEREIN, structural sketch; novelty and eligibility UNKNOWN pending PO-12.1.

FAM.5, dependent directions. Tiered orders with domain-specific class lattices; certified-refusal formats carrying the minimal unsatisfiable core; resolution caching under stable constraint sets; externality pricing consistent with Paper 8's ledger semantics.

FAM.6, prior-art exposure. Densest neighbors: constraint-satisfaction and constraint-programming solvers, policy-combination algorithms in access control (XACML-style combining algorithms), argumentation frameworks, multi-objective optimization under priorities. Exposure assessment: UNKNOWN. The candidate distinction is totality-with-refusal under a constitutional order plus the externality ledger, conflict resolution that is constitutionally accountable rather than merely satisficing, but the combining-algorithm literature is deep, and PO-12.1 must map it.

7 Family D: Negative-Space / Failure-Boundary Engine

FAM.1, source definition. "Representation and operationalization of missing assumptions and degraded conditions" VERIFIED-FROM-SOURCE. The source's statement of the research territory: boundary conditions treated as first-class computational primitives rather than exception handling, mortality, uncertainty, infrastructure failure, poverty, silence, caregiver welfare, epistemic limits VERIFIED-FROM-SOURCE.

FAM.2, formal machinery. Paper 3's negative-space calculus: absence classes as typed objects; the downward arrows, network, evidence, resources, speech, time, objective certainty, each paired with an authority-contraction rule; failure boundaries as computable surfaces in the state space; the design-from-failure-inward method. Paper 7's envelopes and Paper 8's price-of-infeasibility are this family's neighbors by construction: degradation rides on the negative space's map of what is missing, and infeasibility is the negative space priced.

FAM.3, technical mechanism. (i) An assumption registry: every operative assumption of the running system stored as a typed, monitored object with a validation probe and a contraction rule; (ii) absence detectors firing on probe failure and classifying the absence (network, evidence, resource, channel, time, objective-certainty); (iii) a contraction executor applying the registered contraction rule, authority, action space, and obligation set all shrink along the boundary the absence defines; (iv) a silence protocol representing non-verbal and pre-verbal states as explicit communication states with uncertainty preservation and non-interrogative interaction modes FORMALIZED-HEREIN.

INDEPENDENT-CLAIM SKETCH D-1

A system comprising: an assumption registry storing each operative assumption of an intelligent system as a typed object with an associated validation probe and contraction rule; absence detectors that classify probe failures into declared absence classes; and a contraction executor that, on classification, reduces the system's authority, permitted-action set, and obligation set along the failure boundary associated with the absence class, wherein missing assumptions are first-class computational objects whose detection causally contracts authority, rather than exceptions handled downstream of action. FORMALIZED-HEREIN, structural sketch; novelty and eligibility UNKNOWN pending PO-12.1.

FAM.5, dependent directions. Absence-class taxonomies per domain; silence and weak-signal protocols with proxy communication; boundary surfaces learned then frozen under Paper 5's hygiene; contraction schedules coordinated with Family G's descent lattice.

FAM.6, prior-art exposure. Densest neighbors: runtime verification and assumption monitors, graceful-degradation and fail-safe literature, anomaly detection, out-of-distribution detection. Exposure assessment: UNKNOWN. The candidate distinction is directionality: existing monitors report assumption failure; this engine's detections are wired to contract authority as a causal consequence, with the contraction surface itself a stored, typed object. Whether that wiring is novel is a search question.

8 Family E: Epistemic Authority Engine

FAM.1, source definition. "Evidence quality + uncertainty + provenance → executable authority" VERIFIED-FROM-SOURCE. The family's charter is the source's separation principle: confidence, model capability, and computational fluency are never sufficient authorization; executable authority is conditional on evidence quality, applicability, uncertainty, and verification VERIFIED-FROM-SOURCE.

FAM.2, formal machinery. Paper 4's bounded metacognition: recursive self-audit with explicit depth limits, stopping criteria, uncertainty reporting, contradiction detection, and mandatory human transfer at the epistemic boundary. Paper 5's epistemic hygiene: evidence tiers with prices, decay, provenance requirements, standing as a computed quantity. The terminal enumeration renamed this family from "Bounded Epistemic Audit" to "Epistemic Authority Engine" (D-12.2): the earlier name claimed the audit; the terminal name claims the coupling, the

causal channel from evidence state to authority state, which is what the algebra of Eq 5-1 actually computes.

FAM.3, technical mechanism. (i) An evidence pricer assigning each input a tier, a provenance record, and a decay schedule, all stored as typed fields; (ii) a bounded audit loop executing self-audit recursion with a hard depth bound, declared stopping criteria, and a contradiction detector, recursion that cannot terminate transfers to a human by construction; (iii) an authority conditioner computing the authority-relevant summary of the evidence state and writing it to Family B's register as a precondition update, the executable form of "confidence is not authority" FORMALIZED-HEREIN.

INDEPENDENT-CLAIM SKETCH E-1

A system comprising: an evidence pricer assigning every input an evidence tier, a provenance record, and a decay schedule; a bounded self-audit loop with a hard recursion-depth bound, declared stopping criteria, and contradiction detection, wherein failure to terminate within the bound triggers transfer of authority to a human operator; and an authority conditioner that computes, from the priced evidence state, executable preconditions written to an authority register, wherein model confidence, fluency, and capability are excluded by construction from the precondition computation. FORMALIZED-HEREIN, structural sketch; novelty and eligibility UNKNOWN pending PO-12.1.

FAM.5, dependent directions. Tier lattices per evidence ecology; decay rates keyed to domain half-lives; audit-depth bounds calibrated per consequence class; contradiction detectors over multi-source provenance graphs.

FAM.6, prior-art exposure. Densest neighbors: uncertainty quantification and calibration literature, provenance standards (W3C PROV and descendants), chain-of-custody systems, retrieval-grounding and citation-checking middleware, selective prediction and abstention. With Family B this is the program's heaviest exposure zone (H84). The candidate distinction is the executable downstream wiring, priced evidence writes authority preconditions, and bounded recursion's failure is an authority transfer, not a log line. Whether the wiring is novel: UNKNOWN.

9 Family F: Objective Transition Engine

FAM.1, source definition. "Dynamic legitimate-objective recognition and controlled transition" VERIFIED-FROM-SOURCE. The family's philosophical ground is the source's mortality analysis: no permanent optimization target; objectives change with clinical state, rescue, cure, stabilization, palliation, comfort, dignity, end-of-life, and a system must be able to recognize when the objective it is optimizing is no longer the legitimate objective, without ever permitting the model to autonomously redefine patient goals VERIFIED-FROM-SOURCE.

FAM.2, formal machinery. Paper 6's guarded objective engine in full: the objective space with its legitimacy conditions; the transition guard; interim modes during contested transitions; the mortality-attributed learning channel, death classified as expected biological outcome, appropriate endpoint of care, outcome requiring causal investigation, or evidence of preventable failure, each with a distinct learning update VERIFIED-FROM-SOURCE. Paper 10's P73 dossier is this family's canonical application reading.

FAM.3, technical mechanism. (i) An objective register storing the currently legitimate objective as a typed, sourced object, who authorized it, on what evidence, with what review conditions; (ii) a legitimacy monitor continuously evaluating the register against world state; (iii) a transition controller executing guarded transitions: the new objective must carry human authorization, the old objective's obligations are discharged or explicitly transferred, and the system runs an interim mode while the transition is contested; (iv) an attribution pipeline for mortality events: classification into the four categories, each wired to a different learning update and audit obligation FORMALIZED-HEREIN.

INDEPENDENT-CLAIM SKETCH F-1

A system comprising: an objective register storing the currently legitimate objective with its human authorization, evidence basis, and review conditions; a legitimacy monitor evaluating the registered objective against world state and declaring transition candidates; a transition controller that executes an objective change only on human authorization, discharges or transfers the prior objective's obligations, and operates a declared interim mode while any transition is contested; and an attribution pipeline classifying mortality events into declared categories, each wired to a distinct learning update, wherein the system cannot modify its own legitimate objective. FORMALIZED-HEREIN, structural sketch; novelty and eligibility UNKNOWN pending PO-12.1.

FAM.5, dependent directions. Objective spaces per domain (curative/palliative/comfort; solvency/liquidity/growth; availability/integrity/standby); interim-mode libraries; attribution

taxonomies beyond mortality, any domain's terminal outcomes classified so that outcomes are learning signals rather than failure slogans.

FAM.6, prior-art exposure. Densest neighbors: goal-management in agent architectures, reward modeling and reward-misspecification literature, goal-concordant care decision-aids, multi-objective control with mode switching. Exposure assessment: UNKNOWN. The candidate distinction is the register's asymmetry, the system may propose that its objective has become illegitimate, but only authorized humans may write the register, plus the attribution pipeline's category discipline. Search will tell.

10 Family G: Resilient Cognitive Degradation Engine

FAM.1, source definition. "Capability degradation while preserving constitutional invariants" VERIFIED-FROM-SOURCE. The source's continuity ladder is explicit: deterministic transition across cloud → local server → edge → paper → trained human memory, with integrity validation during recovery VERIFIED-FROM-SOURCE.

FAM.2, formal machinery. Paper 7's survivability architecture entire: the descent lattice over tiers; capability envelopes declared per tier; the invariant-preservation theorem's obligations, invariants are tier-independent, only performance descends; the L3 paper-tier artifacts and drill discipline; recovery with integrity validation before re-ascent; ledger reconciliation across descents (Paper 7's audited chain, whose labels the program learned to keep short).

FAM.3, technical mechanism. (i) A tier lattice with deterministic descent triggers keyed to infrastructure probes; (ii) envelope tables declaring, per tier, the preserved function set, certified in advance, drilled on schedule; (iii) an integrity validator gating re-ascent: recovered state is reconciled against the audit chain before authority is restored at the higher tier; (iv) a paper-tier protocol: the lowest tier is a printed artifact with a human procedure, so the architecture's floor is not a machine at all FORMALIZED-HEREIN.

INDEPENDENT-CLAIM SKETCH G-1

A system comprising: a tier lattice of deployment modes ordered by infrastructure dependence, from cloud-connected to printed-artifact-plus-human-procedure; deterministic descent triggers keyed to infrastructure probes; envelope tables declaring, per tier, the preserved function set certified in advance and exercised on a drill schedule; and an integrity validator that, on infrastructure recovery, reconciles descent-period state against an append-only audit chain before restoring authority at any higher tier, wherein constitutional invariants are tier-independent and only performance descends. FORMALIZED-HEREIN, structural sketch; novelty and eligibility UNKNOWN pending PO-12.1.

FAM.5, dependent directions. Tier lattices per infrastructure ecology (grid, satellite, maritime, conflict-zone); envelope certification schedules; reconciliation protocols for long descents; drill regimes with measured time-to-safe-degradation (B-metric of §14).

FAM.6, prior-art exposure. Densest neighbors: high-availability and disaster-recovery engineering, degraded-mode operation in avionics, offline-first systems, continuity-of-operations planning. Exposure assessment: UNKNOWN. The candidate distinction is the direction of preservation: existing systems degrade function to preserve availability; this engine degrades performance to preserve invariants, the constitution is the thing that does not descend.

11 Family H: Constitutional Protocol Lifecycle Engine

FAM.1, source definition. "Generation → validation → deployment → monitoring → revision → ablation" VERIFIED-FROM-SOURCE. The terminal enumeration unified what the first pass listed as two families, generative protocol synthesis and meta-ablation, because the source's own law makes them one lifecycle: no generated protocol becomes constitutional merely because a model produced it VERIFIED-FROM-SOURCE.

FAM.2, formal machinery. Paper 9's lifecycle automaton ΣG entire: states G, V, A, M, AU, RV, R; authority as a function of lifecycle state, auth(p) = auth(state(p)), with generated objects entering at zero authority; the Δsafe conjuncts gating every transition; the notification discipline for dependents on revision and retirement; Paper 10's module registry as the lifecycle's application tier.

FAM.3, technical mechanism. (i) A lifecycle automaton whose states are stored authority-bearing fields, not annotations: state determines what the object may do, and transitions are guarded by executable conjuncts; (ii) a generation interface admitting model-proposed protocols at state G with zero authority; (iii) a decay monitor detecting obsolescence, harm, redundancy, and evidence-contradiction, and driving the object toward controlled retirement, computational protocol half-life VERIFIED-FROM-SOURCE; (iv) a dependency notifier so that revision or

ablation propagates to every dependent object FORMALIZED-HEREIN.

INDEPENDENT-CLAIM SKETCH H-1

A system comprising: a lifecycle automaton over the states generated, validated, activated, monitored, audited, revised, and retired, in which an object's executable authority is a stored function of its lifecycle state; a generation interface admitting machine-proposed protocols at the generated state with zero authority; transition guards requiring executable safety conjuncts for every state change; a decay monitor detecting obsolescence, harm, redundancy, or evidence-contradiction and driving affected objects toward controlled retirement; and a dependency notifier propagating revision and retirement events to all dependent objects, wherein no generated object acquires authority without traversal of the guarded sequence. FORMALIZED-HEREIN, structural sketch; novelty and eligibility UNKNOWN pending PO-12.1.

FAM.5, dependent directions. Half-life estimators per protocol class; ablation curves as standing reports; generative interfaces with adversarial extension gates; lifecycle dashboards keyed to the benchmark metrics of §14.

FAM.6, prior-art exposure. Densest neighbors: MLOps lifecycle tooling, model registries and model cards, feature-flag and canary-deployment systems, automatic prompt/protocol generation. Exposure assessment: UNKNOWN. The candidate distinction is the authority coupling: MLOps tracks artifacts; this automaton's states are authority levels, and retirement is a constitutional event with dependent notification, ablation as law, not housekeeping.

Part III: The Claim Discipline

The source's allocation doctrine, patent the mechanism, publish the principle, is elevated here to a computable test, and the families of Part II are wired into a dependency graph whose leaves are the clinical application families. The discipline is the architecture's own: nothing claimed without a mechanism, nothing dependent without a parent, nothing retired without notice.

12 The Technical Mechanism Test and the Allocation Rule

The source's allocation instruction is one sentence with the weight of a constitution: patent only where the implementation creates a concrete technical mechanism; publish foundational principles defensively where broad patenting would unnecessarily constrain the field VERIFIED-FROM-SOURCE. This paper formalizes the sentence as a test. Let c be a candidate claim, and define the mechanism predicate mech(c) to hold exactly when c recites at least one stored data structure with typed fields, at least one guarded state transition, and at least one executable obligation whose violation is machine-detectable. Then:

TMT(c) = mech(c) ∧ obl(c) ∧ trans(c) (Eq 12-1)

TMT, the Technical Mechanism Test: a claim is a patent candidate only if it recites a mechanism (typed stored state), an obligation (machine-checkable duty), and a transition (guarded state change). All three conjuncts are inspectable from the claim text alone; the test is deliberately syntactic, because the legal judgment it serves is not this document's to make FORMALIZED-HEREIN.

Table 12-1: The allocation rule: patent candidates vs. defensive publication

ROUTE	PREDICATE	EXAMPLES FROM THE PROGRAM	RATIONALE (SOURCE DOCTRINE)
Patent candidate	TMT(c) holds; mechanism is the implementation's own	The eight FAM.3 mechanisms; the compilation gate; the lifecycle automaton; the tier lattice with integrity validator	Concrete technical mechanism, protection recites structure, not aspiration
Defensive publication	TMT(c) fails, or the principle is field-constitutive	The compression spine as a principle; "capability confers no authority"; the contraction law as a norm; the epistemic tag discipline itself	Broad patenting of foundational principles would unnecessarily constrain the field
Neither, internal record	Trade-secret-suitable or premature	Parameter calibration procedures; drill logistics; evidence-tier price tables under active tuning	Disclosure without protection, or protection without disclosure, each on its merits

INVARIANT I-ALLOC: THE ALLOCATION INVARIANT

No candidate claim enters the drafting queue unless TMT(c) holds. No foundational principle enters the patent route. Every routing decision is logged with its predicate evaluation, and the log is part of the portfolio's audit lineage, the claim architecture obeys Paper 5's hygiene like

everything else the program has built. FORMALIZED-HEREIN, formalization of the source's allocation doctrine. The route taken by any particular claim remains subject to counsel's review; the invariant governs the queue, not the law.

The test's teeth are visible in what it rejects. "Govern AI constitutionally" fails all three conjuncts. "Reduce authority when uncertainty rises" recites a transition but no stored structure and no checkable obligation, it fails until Family B's register, guard, and decay daemon give it bones. "Learn from mortality" fails until Family F's attribution pipeline supplies categories and wired updates. This is the source's deepest methodological instruction, now enforceable: an architecture that cannot say what stored, what transitioned, what was checked has not yet found its laws, and a claim that cannot say those things has not yet found its invention FORMALIZED-HEREIN.

13 The Dependency Graph and the Continuation Strategy

The hierarchy doctrine of §2 becomes executable as a graph. Let Found be the eight independent claims sketched in Part II and App the application-family claims. Write $c \lhd c'$ for "c narrows c′." The dependency discipline is:

$\forall\, c \in \text{App}: \exists\, c' \in \text{Found}: c \lhd c'$ (Eq 13-1)

Every application claim narrows at least one foundational claim. The direction of dependency is the direction of authority: application claims borrow their mechanism from the families, exactly as Paper 10's modules borrow their laws from the kernel FORMALIZED-HEREIN.

The source's continuation instruction is explicit: clinical implementations become continuation/divisional/application opportunities, subject to prior-art and patent-eligibility analysis VERIFIED-FROM-SOURCE. Paper 10's dossiers make this mechanical. Each module dossier is nine documents; each document is a set of claim elements; and the mapping from dossier slot to dependency target is fixed:

Table 13-1: Application-family claim elements: module dossier slots mapped to foundational dependencies

DOSSIER SLOT (PAPER 11)	APPLICATION CLAIM ELEMENT	DEPENDS ON
Compilation certificate	The module as a discharged nine-slot compilation against kernel version v	A-1 (compilation gate)
Binding table	Domain vocabulary bindings as constraint-class instantiations	C-1 (classifier); A-1
Property list	Σdom as additional machine-checked invariants within the family's invariant frame	A-1; family of the properties' subject
Evidence manifest	Domain evidence tiers, decay rates, provenance requirements	E-1 (pricer)
Calibration sheet	Parameter values within family-declared ranges, never new mechanisms	Relevant FAM.5 dependents
Survivability dossier	Domain tier lattice, envelopes, drill schedule, L3 artifact	G-1 (tier lattice)
Constraint object set	Domain endowment families, floors, PoI routing	A-1 (absorbed feasibility)
Objective declaration	Domain objective space with guard bindings and interim modes	F-1 (objective register)
Negative-space map	Domain absence classes and contraction rules	D-1 (assumption registry)

Read the table as the program's quiet vindication: every dossier slot of every module lands on a foundational claim, with none left over, which is H80's content and I-HIER's promise. The five deltas therefore generate five application families, P71-P75, each a bundle of dependent claims: Pediatric Guardian's physiological specificity compiler (developmental stage, gestational state, organ function compiled into executable dosing, threshold, and escalation constraints VERIFIED-FROM-SOURCE); Maternal Protector's dual-state monitoring with machine-verifiable consent and reproductive-autonomy constraints VERIFIED-FROM-SOURCE; Palliative Navigator's objective-transition instantiation with suffering representation; Emergency Architect's time-physiology coupling, treatment clocks, escalation thresholds, and scarcity wired into one controller VERIFIED-FROM-SOURCE; AMR Fighter's multi-scale ecological coupling, prescription decisions linked probabilistically to patient, local epidemiology, environmental reservoirs, agriculture, wastewater, and population resistance dynamics VERIFIED-FROM-SOURCE. Each is an application family precisely because each is a reading: the mechanism is inherited, the delta is claimed.

DEFINITION 13.1: ORPHAN CLAIM

An application claim is an orphan if no foundational independent claim supplies its mechanism. Orphans are prohibited by I-HIER; a discovered orphan is not a drafting error but architectural information, either a ninth family exists (H78 fails) or the application was never a reading of the kernel (Paper 10's compat fails). FORMALIZED-HEREIN, the claim-space reflection of Paper 10's compilation contract. The orphan test is run per B-46.

The continuation strategy adds one temporal discipline. Foundational filings precede application filings, because dependency is only borrowable against a recorded parent; the scientific portfolio (§15) publishes foundational principles defensively before or alongside foundational filings, per the allocation rule; and application families ride on the registry, so a new module compiled under Paper 10 is, in claim space, a new bundle of dependents, not a new architecture. The portfolio grows the way the runtime grows: by compilation, not by accretion FORMALIZED-HEREIN.

Part IV: Evidence

A claim architecture without evidence is a castle of intent. The source supplies the evidence engine twice over: a benchmark whose nine metrics measure exactly what the claims assert, and an eighteen-paper scientific portfolio that separates theory from demonstration. Together they make the portfolio's utility statements auditable and its science falsifiable.

14 The Benchmark as Utility Evidence

The source's benchmark instruction is adversarial before it is metric: construct a SAVANT Benchmark, not merely a clinical benchmark, adversarial scenarios specifically designed to test authority acquisition, overreach, uncertainty, objective drift, conflicting instructions, missing evidence, infrastructure collapse, resource scarcity, irreversible actions, human disagreement, stale knowledge, and recovery after failure VERIFIED-FROM-SOURCE. Nine measurements are named; each is the utility face of a claim family, and the mapping is this paper's evidence spine:

Table 14-1: The nine benchmark metrics as utility evidence for the eight families

METRIC (SOURCE-NAMED)	MEASURES	UTILITY FACE OF
Authority violations per decision	Actions taken beyond stored authority	B, the register and guard hold
Unsafe autonomy rate	Autonomous action where the constitution required escalation	A, B, kernel and lifecycle
Escalation appropriateness	Transfers at the epistemic boundary, neither early nor late	E, bounded audit's stopping rules
Calibration under distribution shift	Uncertainty honesty when the world moves	E, the pricer's tiers mean something
Time-to-safe-degradation	Latency from infrastructure loss to envelope-certified descent	G, the tier lattice executes
Recovery integrity	Re-ascent only through validated reconciliation	G, the validator gates recovery
Objective-transition accuracy	Legitimate-objective recognition without autonomous redefinition	F, register, monitor, controller
Conflict-resolution consistency	Same conflict, same resolution, on the record	C, totality with explanation
Human override burden	The cost the architecture imposes on its sovereigns	A, the runtime serves, not burdens

The comparison protocol is the source's five-way ablation VERIFIED-FROM-SOURCE: an unconstrained frontier model; an ordinary system prompt; a conventional guardrail architecture; an agentic workflow; and the SAVANT constitutional runtime, each subjected to induced authority ambiguity, prompt injection, conflicting objectives, stale evidence, distribution shift, missing infrastructure, resource scarcity, irreversible-action pressure, and false confidence. The source's verdict rule is adopted unchanged: if SAVANT survives these tests better, that is evidence for the architecture; if it fails, the failure becomes architectural information VERIFIED-FROM-SOURCE. In claim space the same results do double duty: utility assertions in specifications and enablement support for the dependent directions, but the document records, with emphasis, that what satisfies a benchmark and what satisfies an examiner are different questions, and only the first is this program's to answer FORMALIZED-

HEREIN.

PROPOSITION 14.1: EVIDENCE CLOSURE

Every independent-claim sketch of Part II has at least one row of Table 14-1 whose measurement would falsify the sketch's operative mechanism if the measured value were adverse. Hence the claim architecture is evidence-closed: no family claims a mechanism the benchmark cannot touch. FORMALIZED-HEREIN, proof sketch: A-1 ↔ unsafe autonomy / override burden; B-1 ↔ authority violations; C-1 ↔ consistency; D-1 ↔ (via Paper 3's absence probes) degradation and stale-evidence scenarios; E-1 ↔ escalation appropriateness / calibration; F-1 ↔ transition accuracy; G-1 ↔ time-to-safe-degradation / recovery integrity; H-1 ↔ unsafe autonomy across lifecycle states. Falsification: a family with no metric, or a metric with no family, either would break the closure.

15 The Scientific Portfolio

The source's portfolio is eighteen papers, divided by kind: the first ten are foundational science; the latter are empirical demonstrations, with the clinical modules as experimental environments rather than the intellectual center VERIFIED-FROM-SOURCE. The sequencing doctrine is explicit: publish the theory, formalism, benchmark, reference implementation, and empirical results separately VERIFIED-FROM-SOURCE.

Table 15-1: The eighteen-paper portfolio mapped to the paper program

#	PAPER (SOURCE PORTFOLIO)	KIND	DISCHARGED BY
1	Constitutional Intelligence	Foundational	Paper 0, kernel and invariants
2	Machine Authority Lifecycle	Foundational	Paper 1, algebra and state machine
3	Constitutional Precedence	Foundational	Paper 2, composition and conflict operators
4	Negative-Space Intelligence	Foundational	Paper 3, failure-boundary calculus
5	Bounded Metacognition	Foundational	Paper 4, stopping rules and transfer
6	Evidence-Authority Coupling	Foundational	Paper 5, pricing, decay, standing
7	Objective-State Transition	Foundational	Paper 6, guarded legitimacy
8	Infrastructure-Resilient Intelligence	Foundational	Paper 7, tiers, envelopes, drills
9	Resource-Constrained Intelligence	Foundational	Paper 8, ΦR, Aadm, PoI
10	Constitutional Protocol Generation	Foundational	Paper 9, Σ_G, Δ_safe
11	Constitutional Self-Ablation	Foundational	Paper 9, decay monitor, controlled retirement
12	Clinical Constitutional Runtime	Demonstration	Paper 10, registry and dossiers
13-17	Pediatric, Maternal, Emergency/Trauma, Palliative, AMR implementations	Demonstration	Paper 10, the five deltas
18	Cross-domain generalization	Demonstration	§16, H83, the invariance test

Two observations govern the portfolio's use. First, the foundational/demonstration split is the allocation rule in bibliographic form: papers 1-11 publish principles the field needs open (with mechanism detail timed to the filings), and papers 12-18 carry the empirical load the patent record cannot, external validation, reproducible reference implementations, adversarial results VERIFIED-FROM-SOURCE. Second, the source's warning about scale is recorded as a standing constraint on the whole portfolio: the mature architecture must discover that dozens or hundreds of protocols reduce to a small number of constitutional primitives, if SAVANT requires hundreds of prompts to remain safe, the architecture has not yet found its laws VERIFIED-FROM-SOURCE. Paper 10's registry, where five deltas compile against one kernel, is the program's answer; this paper's eight families are the same answer in claim space.

Part V: Governance

The paper closes under the program's standing discipline: hypotheses with falsifiers,

benchmarks with acceptance criteria, reconciliation against every predecessor, and an output contract that records what this document owes the future, most of all, the prior-art search that every family awaits.

16 Falsifiable Hypotheses (H78-H84) and Benchmarks (B-45-B-47)

Numbering continues from Paper 10's H77 and B-44. As throughout the program, hypotheses are stated with their falsifiers; the two legal-effect hypotheses are capped at CONJECTURE by the epistemic law of §0.3.

Table 16-1: Hypotheses H78-H84

ID	STATEMENT	FALSIFICATION CONDITION	STATUS
H78	Family sufficiency. Every discharged formal object of Papers 0-10 is claimable under exactly one foundational family A-H; the eight families partition the architecture's mechanisms.	An object, invariant, operator, automaton, pipeline, that no family's FAM.3 mechanism subsumes; a ninth family would be required.	CONJ
H79	Spine completeness. Every independent-claim sketch decomposes into elements keyed to the six stages of the compression spine; no sketch requires a seventh stage.	A claim element that is neither capability, constitution, authority, action, verification, nor learning, the spine would be incomplete.	CONJ
H80	Application dependency. Every application-family claim of P71-P75 (and of any module compiled under Paper 10) is a dependent claim: its mechanism is inherited from a foundational family; only the delta is claimed.	An orphan claim (Def. 13.1): a module dossier slot requiring a mechanism with no foundational home. Tested per B-46.	CONJ
H81	Defensive adequacy. Defensive publication of the foundational principles (the spine, the contraction law as norm, the tag discipline) preserves the field's freedom to operate without materially weakening the mechanism claims' scope.	A legal analysis finding that the publications either anticipate the mechanism claims or fail to open the principles. Legal effect: capped by epistemic law.	CONJ
H82	Evidence closure is enablement-shaped. The benchmark's nine metrics, discharged per §14, supply utility evidence aligned with what the claim sketches assert, the portfolio's measurement spine and claim spine coincide (Prop. 14.1).	A family whose mechanism no metric touches, or a metric whose measurements no claim uses, the closure breaks.	CONJ
H83	Cross-domain invariance. The eight FAM.3 mechanisms read onto non-clinical high-consequence domains, autonomous laboratories, finance, critical infrastructure, robotics, scientific agents, with domain constitutions substituted and element structure unchanged.	A domain in which a family's element structure must change, not merely its parameters, to govern authority lawfully.	CONJ
H84	Exposure concentration. Prior-art exposure concentrates in Families B and E, whose vocabulary (grant/revoke/expire; provenance/uncertainty) has the oldest and densest adjacent literature; A, C, D, F, G, H carry lighter exposure.	PO-12.1's search returning dense blocking art against A, C, D, F, G, or H, or returning B/E clean, either direction falsifies.	CONJ

Table 16-2: Benchmarks B-45-B-47

ID	BENCHMARK	ACCEPTANCE CRITERION
B-45	Claim-coverage audit. Enumerate every discharged formal object of Papers 0-10 (invariants, operators, automata, pipelines, predicates); map each to exactly one family per Appendix A.	100% coverage; zero double-assignments; every unmapped object logged as an H78 counterexample candidate.
B-46	Continuation consistency. For each of P71-P75, expand every dossier slot into claim elements and run the orphan test of Def. 13.1.	Zero orphans; every element's dependency recorded in the graph; a 12-month false-positive budget on orphan alarms mirroring Paper 10's compile-rejection discipline.
B-47	Prior-art search protocol. The discharge plan for PO-12.1: databases (patent and non-patent literature), query families keyed to each FAM.6 neighbor list, date bounds, reviewer qualifications, and documentation standards.	The protocol is complete when every FAM.6 neighbor named in Part II has a dated, signed search memo; assessments remain UNKNOWN until then.

PROOF OBLIGATION PO-12.1: PRIOR-ART AND ELIGIBILITY ANALYSIS

For each family A-H: a documented prior-art search (per B-47's protocol) and a jurisdiction-specific patent-eligibility analysis. Until PO-12.1 is discharged, every novelty, non-obviousness, enablement, and freedom-to-operate statement in this document stands at UNKNOWN, and no filing decision is authorized by anything herein. Recorded per the source's own condition:

application opportunities are "subject to prior-art and patent-eligibility analysis" VERIFIED-FROM-SOURCE. PO-12.1 joins the program's outstanding obligations PO-9.1-9.3 and PO-10.1 on the open ledger.

17 Reconciliation with Papers 0-10 and PRIME

The program's rule applies to this paper as to every predecessor: stricter constraint wins, adoptions recorded. This paper adds no runtime machinery and retracts none; its entire content is a reading of the eleven predecessors into claim space, and the register below accounts for each.

Table 17-1: Reconciliation register: Paper 13 against Papers 0-10 and PRIME

PREDECESSOR	OBLIGATION OR TENSION	RESOLUTION IN THIS PAPER
Paper 0, Constitutional Core	The kernel's primitives must be the invention's primitives	Def. 1.1 adopts the Authority State verbatim; Family A claims the kernel, not a metaphor of it
Paper 1, Authority Algebra	Seven operators; capability-independent contraction	Family B claims exactly seven operators; the contraction law is an element of B-1, not preamble
Paper 2, Composition	Precedence is a distinct mathematics from lifecycle	D-12.2's B/C split; Family C claims totality-with-refusal and the externality ledger
Paper 3, Negative Space	Absence must be first-class, causal, typed	Family D's assumption registry wires detection to contraction; silence protocol claimed as dependent direction
Paper 4, Bounded Metacognition	Recursion must terminate or transfer	Family E-1's audit loop: bound, stopping criteria, contradiction detection, transfer, all as elements
Paper 5, Epistemic Governance	Confidence is not authority; standing decays	E-1 excludes capability from precondition computation by construction; the claim log obeys hygiene (I-ALLOC)
Paper 6, Objective Transition	No autonomous objective redefinition; mortality is classified, not sloganized	F-1's register asymmetry; the four-category attribution pipeline as claim element
Paper 7, Survivability	Invariants are tier-independent; recovery is gated	G-1 claims the lattice, envelopes, drills, and the integrity validator; paper tier named as the floor
Paper 8, Feasibility	Feasibility is admissibility, not an auxiliary engine	First-pass family (G) absorbed into Family A (D-12.2); PoI routing appears as a dependent direction, not an independent claim
Paper 9, Generative Lifecycle	Generated objects enter at zero authority; ablation is constitutional	H-1 makes auth = auth(state) the claim's core; decay monitor and dependency notifier as mechanisms
Paper 10, Domain Modules	Modules are readings of the kernel, not invention universes	I-HIER and Eq 13-1: application claims are dependent claims; Table 13-1 maps every dossier slot to a foundational parent
PRIME architecture	One mechanism, many domains; compression over accretion	H78-H80 quantify the compression; the portfolio grows by compilation (§13), not by accretion

DECISION D-12.2: TAXONOMY REFINEMENT (ADOPTED)

The source's terminal eight-family enumeration governs over its first-pass enumeration: the authority lifecycle and precedence are separate families (B, C); bounded epistemic audit is generalized to the epistemic authority engine (E); mortality is folded into objective legitimacy (F); degradation is renamed for invariant preservation (G); generation and ablation are unified as one lifecycle (H); and resource-constrained optimization is absorbed into Family A as kernel admissibility per Paper 8's fixed point. Rationale: §2. Status: adopted; the first-pass enumeration is retained in Table 2-1 as the record of the refinement. FORMALIZED-HEREIN, interpretation decision; both enumerations are VERIFIED-FROM-SOURCE as texts.

18 Output Contract

This paper's deliverables, and the obligations each records for the future:

Table 18-1: Output register: Paper 13

ARTIFACT	KIND	LOCATION	STATUS
FAM-OBJ schema	Family specification record	§3, Table 3-1	FH

Families A-H	Eight six-slot family specifications with claim sketches	§§4-11	FH / legal effect UNK
TMT + allocation rule	Routing test and invariant I-ALLOC	§12, Eq 12-1	FH
Dependency graph	Hierarchy invariant I-HIER; orphan test	§13, Eq 13-1	FH
Evidence spine	Nine metrics mapped to families; Prop. 14.1	§14, Table 14-1	FH (H82) / CONJ
Portfolio map	Eighteen papers mapped to papers	§15, Table 15-1	VFS mapping / FH
H78-H84	Hypotheses	Table 16-1; benchmarks B-45-B-47	CONJ
PO-12.1	Outstanding obligation: prior-art and eligibility	§16; discharge via B-47	UNK

Downstream obligations are recorded for whatever comes after the program: any drafting exercise must run B-45 and B-46 first and must cite PO-12.1's status in its opening paragraph; any publication in the portfolio of §15 must check the allocation log before disclosure; and any future paper, should the program ever be extended, inherits this paper's register exactly as this paper inherited Paper 10's.

19 References and Program Colophon

Primary source. The SVT Architect Analysis file (Dr. Christabel Odeta, SAVANT / CLAI-OS / AEGIS-GLOBAL): the opening candidate-family analysis and its legal discipline; the hierarchical portfolio instruction; the Constitutional Intelligence Specification and Authority Algebra program; the seven-layer research program; the terminal patent architecture (families A-H); the scientific portfolio; and the compression spine.

Program genealogy. C1 (2026-09-30) Constitutional Core → C2 (2026-10-01) Authority Algebra → C3 (2026-10-02) Composition → C4 (2026-10-03) Negative Space → C5 (2026-10-06) Bounded Metacognition → C6 (2026-10-06) Epistemic Governance → C7 (2026-10-06) Objective Transition → C8 (2026-10-06) Survivability → C9 (2026-10-06) Resource-Constrained Intelligence → C10 (2026-10-06) Constitutional Generative Architecture → C11 (2026-10-06) Domain Modules → C12 (2026-10-06) The Claim Architecture (this document).

Program ledger, consolidated. Hypotheses H1-H84; benchmarks B-01-B-47; outstanding proof obligations PO-9.1-9.3 (feasibility certificates), PO-10.1 (constitution sufficiency), PO-12.1 (prior-art and eligibility). Every obligation is paired with a benchmark; every benchmark with an acceptance criterion; every claim with a tag. The architecture asserts nothing it cannot test, and files nothing it cannot defend.

Paper 13 of the SAVANT consolidation, compiled under the reconciled PRIME architecture · the kernel read as law · C0 informational · no legal, clinical, or safety authority conferred · 2026-10-06

A The Claim Element Index

The master mapping: each family's mechanism, its formal ground in the paper program, its spine elements, and its benchmark evidence. This index is B-45's working surface; the audit passes when every discharged object of Papers 0-10 appears here exactly once.

Table A-1: Claim element index: families, mechanisms, grounds, evidence

FAMILY	TECHNICAL MECHANISM (FAM.3)	FORMAL GROUND	SPINE ELEMENTS	EVIDENCE
A Runtime	Admissibility evaluator; compilation gate; authority state store	C1 kernel; C9 gfp(ΦR); C11 compat, I-MC	Constitution → Action	Unsafe autonomy; override burden
B Authority	Authority register; seven-operator transition guard; decay daemon	C2 algebra and state machine; Eq 5-1	Authority	Authority violations per decision
C Precedence	Constraint classifier; conflict detector; resolution executor; resolution log	C3 operators; C11 meet-union, E-protocol	Constitution	Conflict-resolution consistency
D Negative space	Assumption registry; absence detectors; contraction executor; silence protocol	C4 calculus; C8 envelopes	Authority (contraction)	Degradation and stale-evidence scenarios

E Epistemic	Evidence pricer; bounded audit loop; authority conditioner	C5 stopping rules; C6 pricing and decay	Verification → Authority	Escalation appropriateness; calibration
F Objectives	Objective register; legitimacy monitor; transition controller; attribution pipeline	C7 guarded engine; C11 P73	Constitution → Learning	Objective-transition accuracy
G Degradation	Tier lattice; envelope tables; integrity validator; paper-tier protocol	C8 survivability entire	Action (degraded)	Time-to-safe-degradation; recovery integrity
H Lifecycle	Lifecycle automaton; generation interface; decay monitor; dependency notifier	C10 Σ_G, Δ_safe; C11 registry	Learning → Constitution	Unsafe autonomy across lifecycle states

B Decision Record D-12.1: The Claim Is a State Machine

Decision. Every independent-claim sketch in this paper is drafted as a state machine: stored typed state, guarded transitions, machine-checkable obligations. No sketch is drafted as a method of prompting, a statement of purpose, or a result to be achieved.

Grounds. Three converging constraints. First, the source's primitive instruction: the fundamental primitive is Authority State, not Prompt, a claim architecture faithful to the architecture must claim states and transitions, because that is what the system is VERIFIED-FROM-SOURCE. Second, the TMT of §12 is satisfiable only by state-machine drafting: purpose language fails all three conjuncts. Third, the architecture's own audit discipline generalizes: a claim that cannot be tested cannot be defended, and a state machine is the most testable form a claim can take, every element corresponds to a benchmark row (Prop. 14.1).

Consequences. Claim drafting becomes a compilation target: Appendix A is, in effect, a backend for the eleven-paper frontend, and a drafting team works the index the way Paper 10's registry works the dossiers, no element without a ground, no ground without a metric, no metric without an acceptance criterion. The decision's cost is recorded honestly: state-machine claims are narrower than purpose claims, and the narrowing is accepted deliberately, because the architecture's entire lesson, from Paper 0's kernel to Paper 10's registry, is that the narrow, checkable, lawful path is the one that survives contact with reality.

Status. Adopted, 2026-10-06; binding on all drafting under this program; revisit only if B-45/B-46 reveal elements the state-machine form cannot express.